Associations of Sleep Duration, Sleep Debt, and Social Jetlag with Self-Reported Cancer History among U.S. adults: a NHANES 2017–March 2020 Analysis

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Abstract

Background: Sleep and circadian disruption have been proposed as modifiable factors relevant to cancer, but epidemiologic evidence remains inconsistent.

Objective: To examine associations of sleep duration, sleep debt, and sleep-corrected social jetlag with self-reported cancer history among U.S. adults.

Methods: This cross-sectional study used NHANES 2017 to March 2020 pre-pandemic data. Adults aged 20 years or older with non-missing cancer history were included (n = 9,141). Sleep measures and cancer diagnosis were self-reported. Survey-weighted logistic regression models were fit separately for each sleep exposure and adjusted for demographic, socioeconomic, lifestyle, and health-related covariates.

Results: After adjustment for confounders, sleep duration, sleep debt, and sleep-corrected social jetlag were not associated with self-reported cancer history.

Conclusion: Findings did not support clear associations between the sleep measures evaluated and prevalent self-reported cancer history in this cross-sectional NHANES sample.

Introduction

Background:

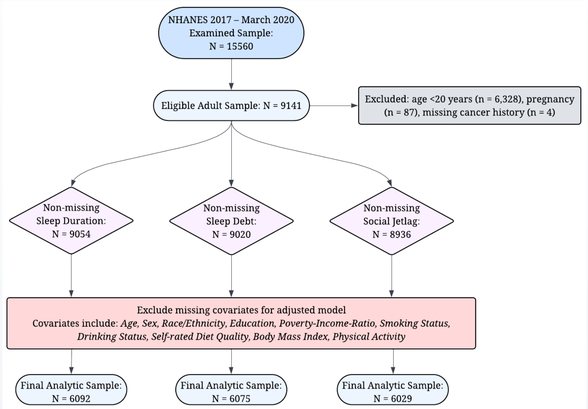
- Sleep and circadian health may influence cancer-related pathways through inflammation, immune function, hormonal regulation, and circadian timing.
- Sleep duration-cancer evidence inconsistent; sleep debt and social jetlag evidence limited.
- Few population-based studies of multiple sleep dimensions and cancer history.

Study Aim:

Examine associations of sleep duration, sleep debt, and sleep-corrected social jetlag with self-reported cancer history among U.S. adults in NHANES 2017 to March 2020.

Methodology

Figure 1. Analytic Sample Construction



Measures

- *All outcome and sleep exposures were self-reported.*
- Outcome: Cancer or malignancy diagnosed by a health professional.
- Sleep duration: Usual weekday sleep; <7, 7 to 9, or >9 hours.
- Sleep debt: Weekend minus weekday sleep; ≤0.5, >0.5 to <2, or ≥2 hours.
- Sleep-corrected social jetlag: Absolute weekday-weekend sleep timing difference; per 1-hour increase.

Statistical analysis: Survey-weighted logistic regression estimated OR and 95% CI, accounting for survey weights, strata, and PSU. Separate models were fit for each exposure:

- Model 1: Unadjusted.
- Model 2: Model 1 + age, sex, race and ethnicity, education, and poverty-income ratio.
- Model 3: Model 2 + BMI, smoking status, alcohol use, physical activity, and diet quality.

Sensitivity analyses: additional shift adjustment and multiple imputation

Results

Table 1. Selected survey-weighted characteristics of the analytic sample

Characteristic	Overall (N=6027)
Age, years (mean ± SE)	48.7 (0.7)
Sex, %	
Male	3,074 (49.7%)
Female	2,953 (50.3%)
Race and ethnicity	
Non-Hispanic White, %	2,343 (66.9%)
Non-Hispanic Black, %	1,554 (10.3%)
Hispanic, %	1,269 (14.5%)
Other race and ethnicity, %	861 (8.4%)
Educational attainment	
Less than high school, %	935 (9.0%)
High school graduate or GED, %	1,451 (26.6%)
Some college or associate degree, %	2,102 (31.8%)
College graduate or above, %	1,539 (32.6%)
Family poverty-to-income ratio	
Less than 1.3, %	1,596 (17.3%)
1.3 to less than 3.0, %	2,042 (27.8%)
3.0 or greater, %	2,389 (55.0%)

Table 2. Survey-weighted associations of sleep exposures with self-reported cancer history

Sleep exposure and contrasts	Model 1	Model 2	Model 3
	Odds Ratio (95% Confidence Interval)		
Sleep duration (hours)			
<7 vs. 7 to 9	0.8 (0.61, 1.05)	0.98 (0.70, 1.38)	0.93 (0.57, 1.53)
>9 vs. 7 to 9	1.29 (0.94, 1.76)	1.14 (0.75, 1.76)	1.10 (0.58, 2.09)
Sleep debt (hours)			
>0.5 to <2 vs. ≤0.5	0.56 (0.39, 0.71)*	0.85 (0.54, 1.32)	0.85 (0.44, 1.65)
≥2 vs. ≤0.5	0.37 (0.27, 0.50)*	0.81 (0.54, 1.24)	0.83 (0.45, 1.53)
Sleep-corrected social jetlag			
Per 1-hour increment	0.52 (0.41, 0.68)*	0.86 (0.74, 1.01)	0.86 (0.70, 1.06)

Results Summary

- No clear adjusted associations between sleep measures and self-reported cancer history.
- Similar conclusions after shift adjustment and multiple imputation.

Discussion

Public Health Relevance

- Sleep concerns were common: short sleep 24.2%, sleep debt >0.5 hours 39.3%, mean social jetlag 0.7 hours.
- Self-reported cancer history affected 11.9% of adults in the analytic sample.
- Despite null adjusted associations, sleep and circadian health remain important for chronic disease prevention, quality of life, and survivorship care.

Strengths

- Nationally representative U.S. adult sample.
- Three sleep dimensions evaluated.
- Survey-weighted analyses accounted for NHANES design.
- Sensitivity analyses included shift adjustment and multiple imputation.

Limitations

- Cross-sectional design; reverse causation possible.
- Prevalent, self-reported cancer history; potential survivorship and prevalence-incidence bias.
- Combined cancer sites; no diagnosis timing, stage, treatment, recurrence, or severity.
- Self-reported sleep; limited long-term sleep pattern capture.
- Current shift schedule only; no lifetime shift-work history.
- Possible residual confounding.

Future Directions

- Clarify whether sleep and circadian disruption are associated with incident, site-specific cancer risk.
- Prospective studies with cancer diagnosis after sleep assessment.
- Objective or repeated sleep/circadian measures.
- Assess chronotype and lifetime shift-work history.

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